

SIH26034 — Architecture, API & Data Model v2

FINAL REFERENCE PACK v2 — audited against the current SIH26034 blueprint/build prompts and current official-source requirements.

Stack

Next.js App Router/TypeScript/Tailwind/Lucide; FastAPI/Python/Pydantic; OpenCV; PaddleOCR; constrained structured extraction; SQLite/WAL for MVP; ReportLab; ORM boundary for later migration.

Pipeline

Create case → upload → server hash → quality gate → calibration/OCR → structured extraction → normalization/sanity gate → deterministic rules → findings → correction → re-analysis → finalise → report → verification.

Evidence

Front/back mandatory; side/base optional. Each evidence item has role, sequence, server receipt time, hash, quality verdict and immutable original. Retakes supersede working images without deleting history.

Provenance

Every field stores source image, source text span/coordinates, confidence and machine/officer origin. Corrections are audited.

Audit/time

State changes, corrections and report events create hash-chained audit entries. Server receipt time is authoritative; client time may be retained separately. Do not call the chain tamper-proof.

Verification/API

Freeze the API contract early. Include authentication/session, inspection lifecycle, evidence upload, analysis, corrections, reports, history/case retrieval, health and unauthenticated report verification.

Fallback

Model/network failure must degrade gracefully. Offline sync is P2; deterministic/manual paths remain available where defined.